

PcaAD

```
**Type:** Statistical / ML-based  
**Library:** ADTK  
**Owner:** Joshua Stackpole  
**Status:** Complete
```

What It Does

PcaAD applies Principal Component Analysis to multivariate sensor data and flags timestamps where the reconstruction error exceeds a learned threshold. It detects anomalies where the normal relationships between sensors (s1, s2, s3) break down simultaneously rather than flagging individual sensor spikes.

Input / Output

| | Format |

|---|---|

| Input | Pandas DataFrame with DatetimeIndex, float values, columns: s1, s2, s3 (normalised to [0,1]) |

| Output | Dict — anomaly_flag (pd.Series bool), model_name (str), timestamp (DatetimeIndex) |

How to Run

```
```python  
From data_science/
from detectors.adtk_pcaad import PcaADDetector
from preprocessor import load_and_prepare

df, scaler = load_and_prepare("datasets/complex.csv")
detector = PcaADDetector(k=2)
result = detector.detect(df)
print(result["anomaly_flag"].sum(), "anomalies detected")
```
```

Or via the pipeline:

```
```bash  
cd data_science
python pipeline.py --benchmark
```
```

Dependencies